Aniket Shinde

Applied AI / ML Engineer

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Profile

AI/ML engineer focused on building and evaluating intelligent systems. My work includes retrieval-augmented generation (RAG), NLP pipelines, fine-tuned language and speech models, and continual-learning systems, with an emphasis on rigorous evaluation, experimentation, and production-oriented design.

Technical Skills

AI / ML: Python, NumPy, Pandas, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, SBERT, fine-tuning, continual learning
LLMs & Agents: Retrieval-Augmented Generation (RAG), hybrid retrieval, LangGraph, LiteLLM, Gemini, Llama, Model Context Protocol (MCP)
Backend & Data: FastAPI, SQL, PostgreSQL, pgvector, Redis, Celery, MinIO/S3
MLOps & Infrastructure: Docker, MLflow, GitHub Actions, pytest
Frontend & AI Development Tools: React, TypeScript, Codex, Claude Code

Projects

Legal Document Citation RAG **2026**

- Built an end-to-end multi-tenant legal-document Q&A platform with FastAPI, async SQLAlchemy, Celery, Redis, MinIO, PostgreSQL/pgvector, React/TypeScript, Docker, and Gemini 2.5 Flash.
- Implemented hybrid retrieval by fusing pgvector cosine search with PostgreSQL full-text search through Reciprocal Rank Fusion; answers return filename, page, section, and verbatim snippet metadata for traceability.
- Built deterministic CUAD retrieval evaluation for Hit@k/MRR with MLflow logging; 512-token chunks improved Hit@6 from 0.194 to 0.224 over 256-token chunks.

Call Compliance Audio Intelligence Platform **2026**

- Led a 4-person Sheridan capstone team building a call-compliance audio intelligence platform spanning ASR, emotion modeling, LLM rubric evaluation, backend ingestion, and React playback.
- Designed the LLM call-evaluation pipeline: deterministic transcript context, Pydantic-validated rubric extraction, timestamped evidence anchoring, chapter segmentation, and audio-required modality flags.
- Validated speech-processing inputs for the rubric: per-channel faster-whisper achieved 90.2% normalized accuracy across 22 calls / 32,059 words; the team Wav2Vec2 emotion model reached 83.41% 6-class accuracy.

AV Technology Patent Mining **2025**

- Built an NLP pipeline over 667 US autonomous-vehicle patents using SBERT embeddings, UMAP, HDBSCAN, TF-IDF labeling, Isolation Forest outlier detection, Prophet forecasting, and Gemini structured extraction.
- Surfaced semantic clusters, filing trajectories, and embedding-space outliers across the patent corpus to map autonomous-vehicle technology trends.

Catastrophic Forgetting Mitigation Framework **2025**

- Built a unified PyTorch continual-learning benchmark implementing EWC, Synaptic Intelligence, PackNet, and Experience Replay across Task-Incremental and Class-Incremental settings.
- Recovered a collapsing Class-IL baseline from 19% to 37% cumulative accuracy with reservoir-sampled Experience Replay, outperforming naive FIFO rehearsal at 32%.

Experience

Sheridan College (FAST) **December 2025 – Present**
Graduate Research Assistant *Oakville, ON*

- Primary technical author on a faculty-supervised manuscript prepared for IEEE IT Professional on enterprise humanoid-robotics adoption; led the evidence synthesis, paper structure, and technical narrative across 50+ academic, industry, and deployment sources.
- Produced 10+ publication-ready figures and visualizations to IEEE print standards, translating research findings into clear diagrams supporting governance, deployment, and adoption analysis.

Education

Sheridan College - Trafalgar **December 2026**
Postgraduate Certificate, Artificial Intelligence (GPA: 3.84/4.0) *Oakville, ON*
Vellore Institute of Technology - Bangalore **June 2025**
Postgraduate Certificate, Artificial Intelligence (GPA: 9.9/10.0) *Bengaluru, KA*
University of Illinois at Chicago **May 2024**
Bachelor of Science, Computer Science *Chicago, IL*